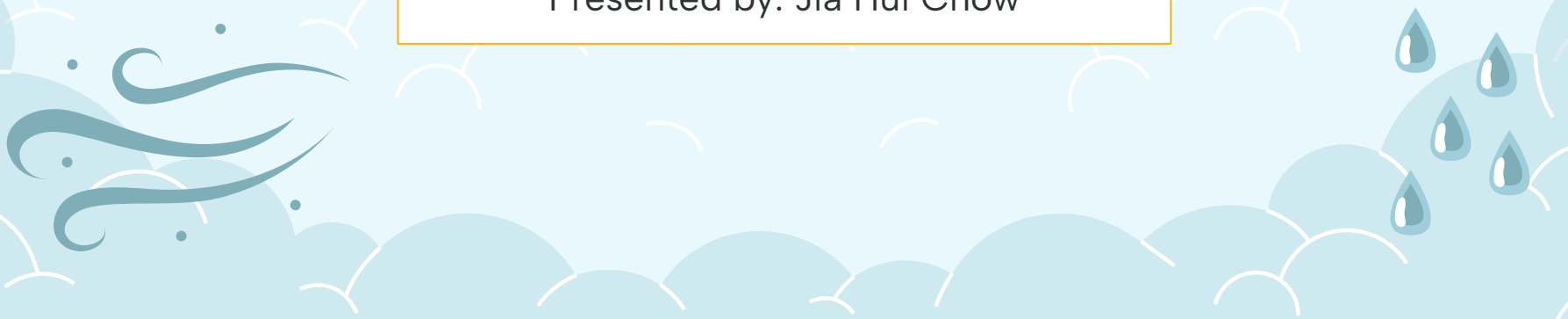




DATS 2102

50 Cities, 365 Days: A Global Story of Weather, Climate, and Extremes in 2025

Presented by: Jia Hui Chow





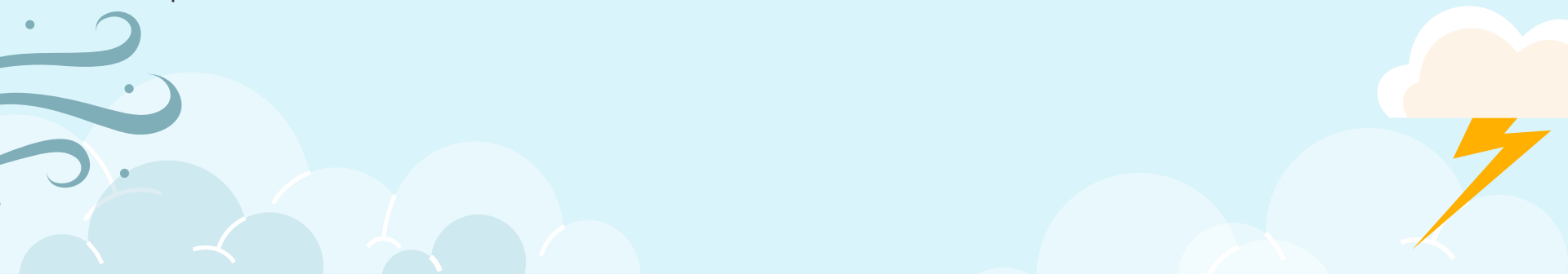
Why compare weather across cities?

Weather is universal, but it is not experienced in the same way everywhere.

A hot day in **Dubai**, a humid day in **Singapore**, and a summer day in **New York** may all seem similar in one number, but they reflect very different climate realities.

Research Question

How do major global cities differ in their weather profiles across 2025, and what do **temperature**, **rainfall**, **humidity**, **seasonality**, and **extreme events** reveal about urban climate experience?



Data and analytical workflow

Data



Dataset

- **Global Daily Weather Data 2025**
- **18,250 observations** = 365 days × 50 cities
- Source: **Open-Meteo Historical Weather API**

Variables

- temperature: max, min, mean
- precipitation and rain
- wind speed
- Humidity
- latitude / longitude

Workflow

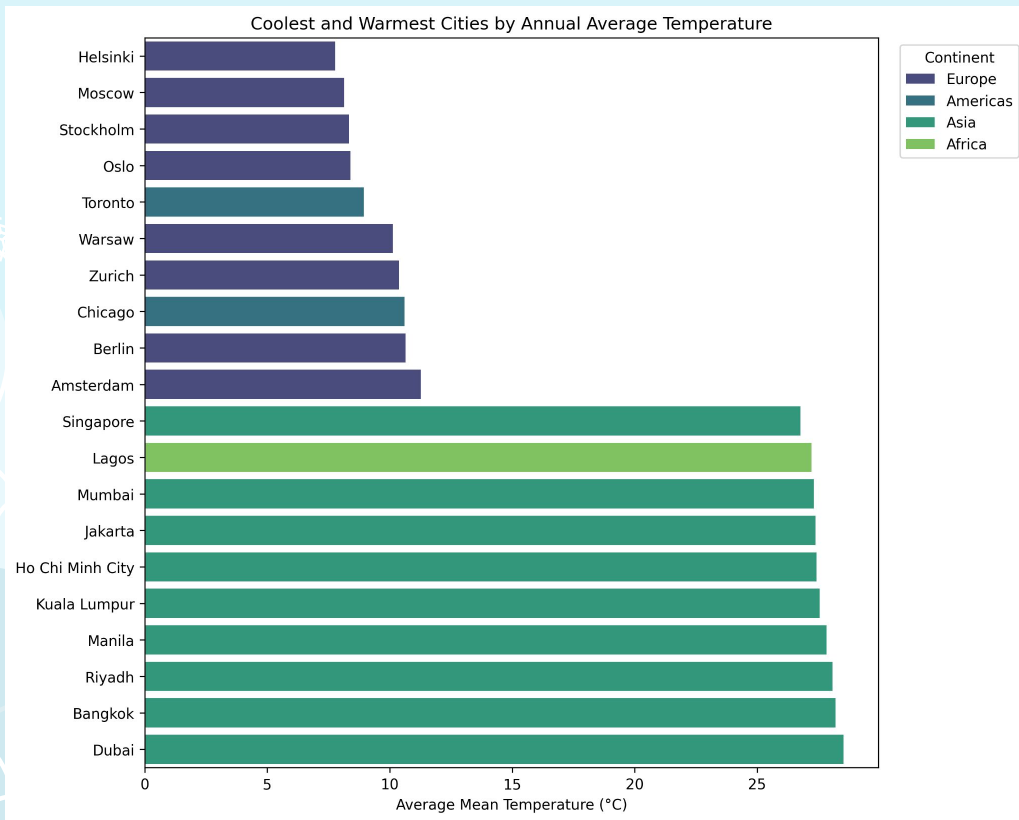


Wrangling and preparation

- converted dates and extracted month / quarter
- created *temp_range_c*
- defined:
 - *extreme_heat_day*
 - *heavy_rain_day*
 - *high_wind_day*
- merged cities with continent labels
- built:
 - city-year summaries
 - monthly summaries
 - country summaries for maps

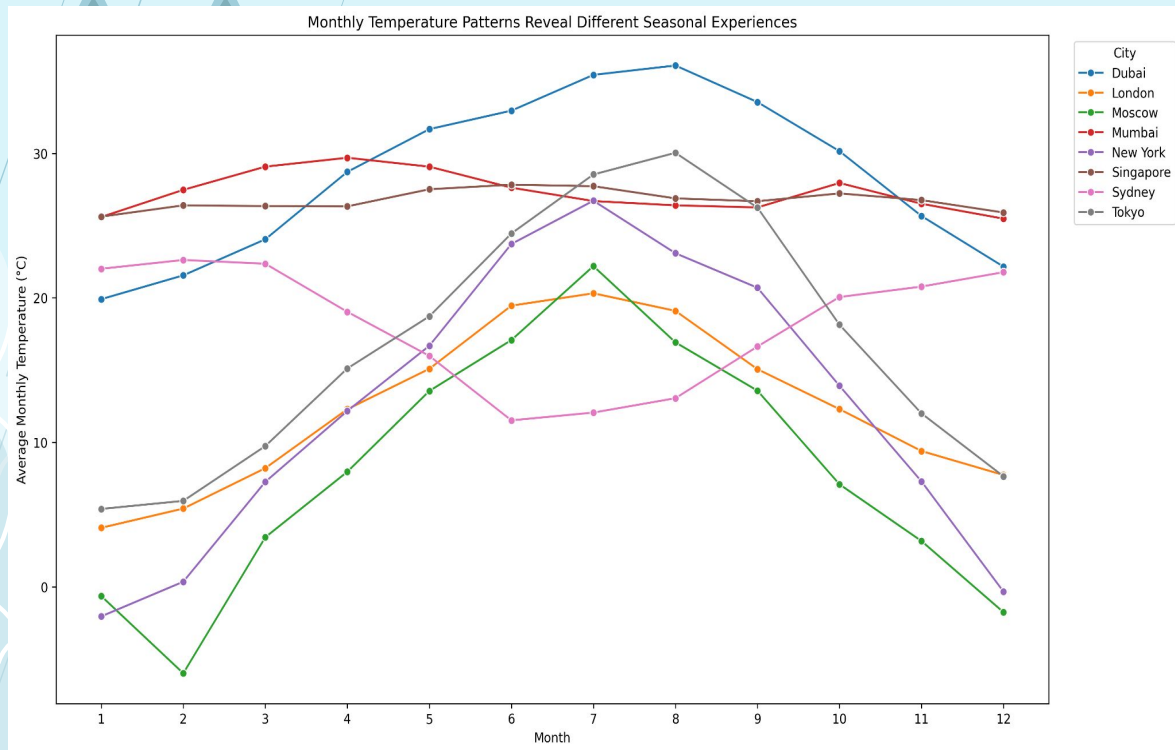


The coolest and warmest cities are easy to spot



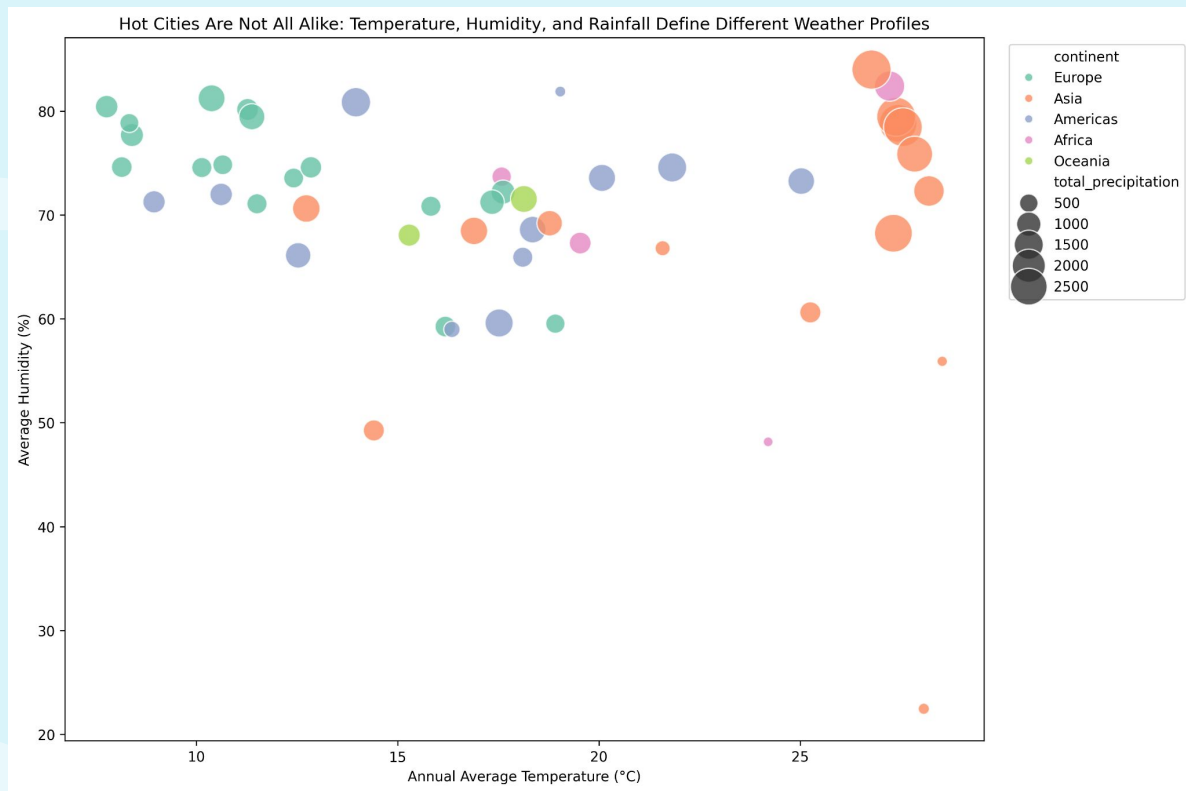
- The **coolest cities** are concentrated in **Northern Europe** and **North America**
- The **warmest cities** are concentrated in **Asia, Africa,** and the **Middle East**
- Cities such as **Helsinki, Moscow,** and **Stockholm** sit at the cool end
- Cities such as **Dubai, Bangkok,** and **Riyadh** sit at the warm end

Cities with similar averages can feel completely different over time



- **Singapore** stays warm and stable all year
- **Moscow** and **New York** show strong seasonal swings
- **Dubai** is warm year-round but peaks sharply in summer
- **Sydney** reverses the Northern Hemisphere seasonal pattern

Hot cities are not all alike

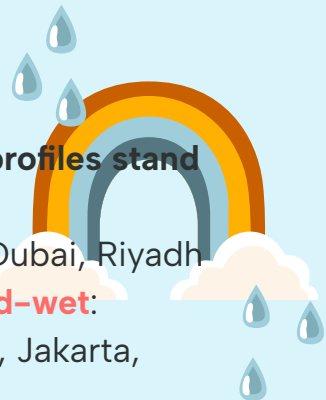


Three weather profiles stand out

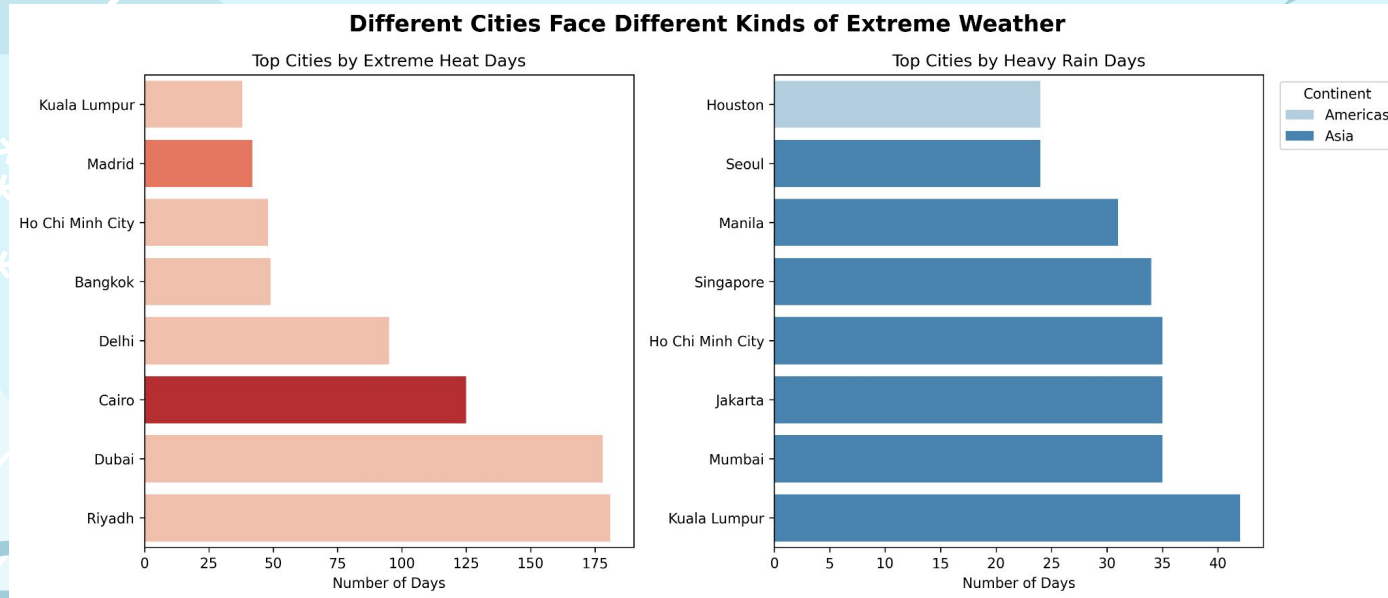
- **Hot-dry:** Dubai, Riyadh
- **Hot-humid-wet:** Singapore, Jakarta, Manila
- **Cooler but humid:** many European cities

Why it matters

- dry heat and humid heat are experienced differently
- rainfall adds another layer to urban weather conditions

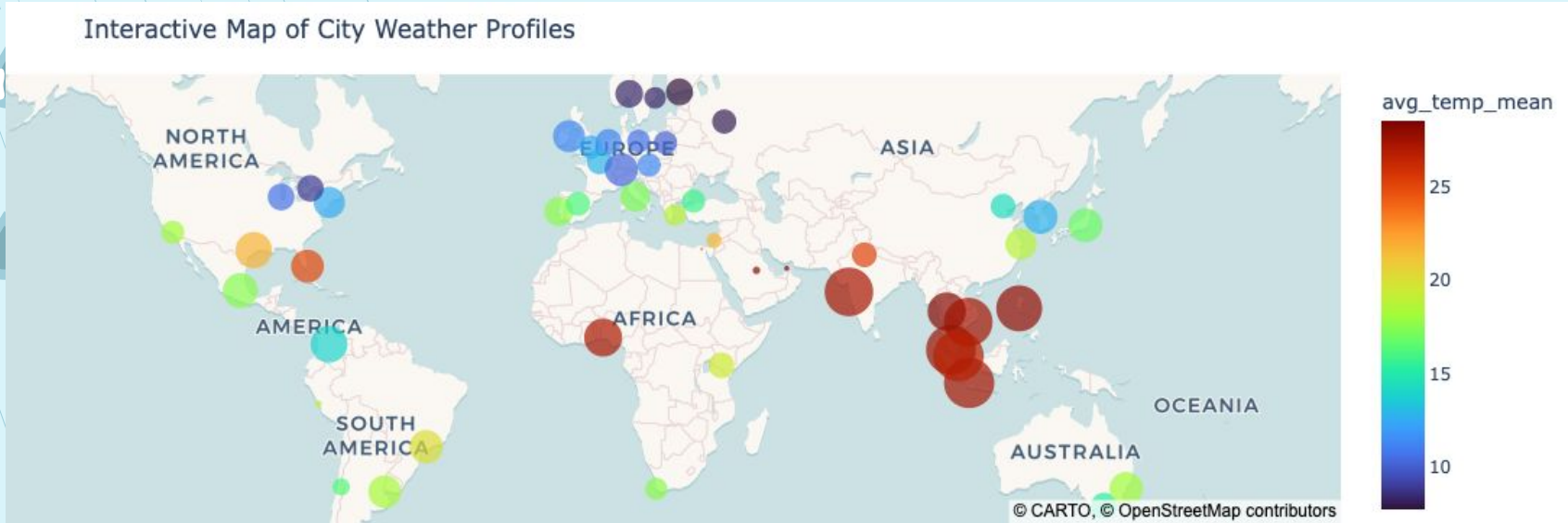


Different cities face different kinds of extreme weather

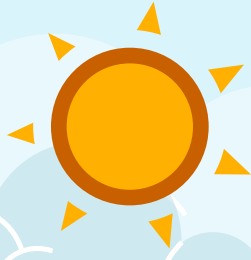


- **Extreme heat** is concentrated in cities such as **Riyadh, Dubai, Cairo, and Delhi**
- **Heavy rain** is concentrated in cities such as **Kuala Lumpur, Singapore, Jakarta, Mumbai, and Ho Chi Minh City**
- The cities most exposed to heat are not the same as those most exposed to intense rainfall

These extreme patterns are geographically structured



- **Middle Eastern cities** cluster as **hot-dry** profiles
- **South and Southeast Asian cities** cluster as **hot-humid-wet** profiles
- **European and North American cities** tend to be cooler and more seasonal



From insights to prediction and interpretation

ML

Predicting extreme heat days

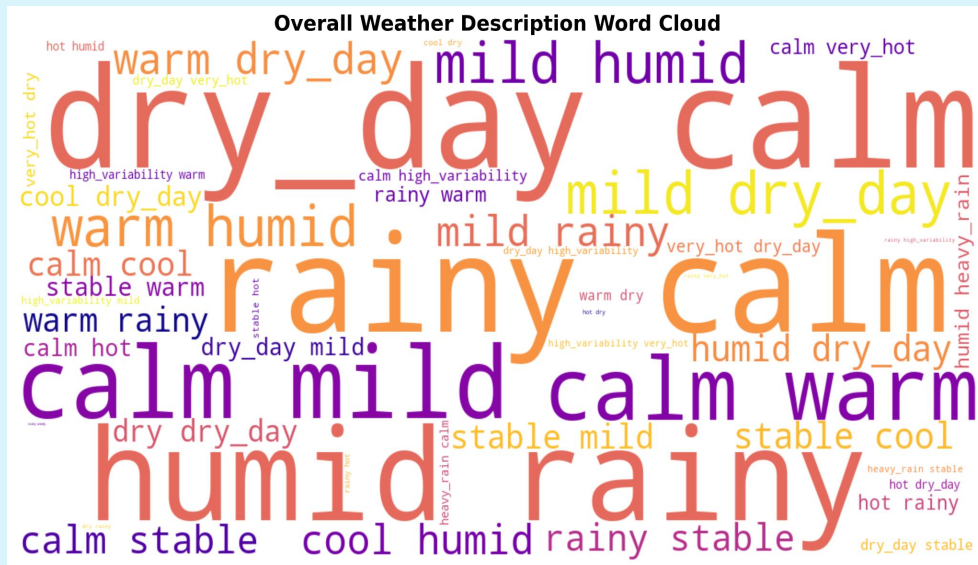
- Model: **Random Forest classifier**
- Target: *extreme_heat_day*
- Strong performance:
 - **Accuracy: 0.99**
 - **ROC AUC: 0.997**

Most important features

- *temp_mean_c*
- *temp_min_c*
- *humidity_pct*
- *latitude / longitude*
- *month*



NLP-style visualization



Limitations and next steps

Limitations

- covers only **one year (2025)**
- includes only **50 cities**
- city-level data hides within-city variation
- country maps reflect **sampled cities**, not full national climate
- extreme thresholds are analytical choices

Next steps

- extend across multiple years
- include more cities and regions
- add **heat index / feels-like temperature**
- build a fuller interactive dashboard
- explore more advanced climate clustering

Final insight

Urban weather is not defined by temperature alone — it is shaped by the interaction of heat, humidity, rainfall, seasonality, and extremes.

- cities with similar averages can have very different weather experiences
- geography and seasonality structure these differences
- the result is a set of distinct urban weather identities

Weather is not one global story — it is many local stories shaped by place.



**THANK
YOU!**

